11+ Maths Practice Paper

Tiffin-style questions | 10 problems | Suggested time: 30 minutes

Name: _____ Date: _____ Score: _____ / 10

Answer every question. Show your working in the box, then write your final answer on the line. Calculators are not allowed. Questions 1 to 5 are warm-ups; questions 6 to 10 are harder.

Question 1 - Fractions

3/8 of a number is 27. What is 5/6 of the same number?

Answer: _____

Question 2 - Ratio

Amy and Ben share some sweets in the ratio 4 : 7. Ben gets 15 more sweets than Amy. How many sweets are there altogether?

Answer: _____

Question 3 - Sequences

Here is a sequence: 7, 11, 15, 19, 23, ...
What is the 50th term?

Answer: _____

Question 4 - Percentages

A coat costs £80. Its price goes up by 25%. Later, in a sale, the new price is cut by 20%. What is the final price of the coat?

Answer: _____

Question 5 - Perimeter and area

A rectangle has a perimeter of 46 cm. Its length is 5 cm more than its width. What is its area?

Answer: _____

Question 6 - Speed

A car travels 120 km at 60 km/h. It then travels another 120 km at 40 km/h. What is its average speed for the whole journey?

Answer: _____

Question 7 - Averages

The mean of five numbers is 14. One number is taken away, and the mean of the four numbers left is 12. Which number was taken away?

Answer: _____

Question 8 - Angles

One angle of an isosceles triangle is 40°. Give **both** possible pairs of sizes for the other two angles.

Answer: _____

Question 9 - Number puzzle

A number leaves a remainder of 1 when it is divided by 2, 3, 4, 5 or 6. It is also exactly divisible by 7. What is the smallest number that does this?

Answer: _____

Question 10 - Problem solving

Pens cost 35p and pencils cost 20p. Ryan spends exactly £2.50 and buys at least one of each. How many different combinations of pens and pencils could he have bought? List them.

Answer: _____

Answers and Working

Question 1 - Fractions

60. 1/8 of the number is 27 ÷ 3 = 9, so the number is 72. Then 5/6 of 72 = 60.

Question 2 - Ratio

55. 7 - 4 = 3 parts = 15, so 1 part = 5. The total is 11 parts × 5 = 55.

Question 3 - Sequences

203. The terms go up by 4 each time, so the nth term is 4n + 3. The 50th term is 4 × 50 + 3 = 203.

Question 4 - Percentages

£80. £80 × 1.25 = £100, then £100 × 0.8 = £80. *Trap:* +25% followed by -20% is not "+5%".

Question 5 - Perimeter and area

126 cm². Width + length = 23, and the length is the width + 5. So the width is 9 cm and the length is 14 cm, giving 9 × 14 = 126.

Question 6 - Speed

48 km/h. The journey takes 2 h + 3 h = 5 h for 240 km, and 240 ÷ 5 = 48. *Trap:* the answer is not 50, the average of the two speeds.

Question 7 - Averages

22. The five numbers add up to 5 × 14 = 70. The four left add up to 4 × 12 = 48. So the number taken away is 70 - 48 = 22.

Question 8 - Angles

70° and 70°, or 40° and 100°. If 40° is the top angle, the other two are each (180 - 40) ÷ 2 = 70°. If 40° is one of the two equal angles, the angles are 40°, 40° and 100°.

Question 9 - Number puzzle

301. The lowest common multiple of 2 to 6 is 60, so the number is 60k + 1. Checking in turn: 61, 121, 181 and 241 don't divide by 7, but 301 = 7 × 43 does.

Question 10 - Problem solving

2 ways: 2 pens + 9 pencils (70p + £1.80), or 6 pens + 2 pencils (£2.10 + 40p). The number of pens has to be even to make a whole-pound total.